

Assignment-3

Task 1:

(a) - support for $\{\text{Classic Poster}\} = 4$

- support for $\{\text{Rubik's Cube, Spices Pack}\} = 2$

- support for $\{\text{Box of Chocolates, Sci-Fi Book, Horror Novel}\} = 0$

(b) Let's denote:

$\{\text{Sci-Fi Book, Wireless Mouse}\} : A$

$\{\text{Laptop}\} : B$

$\{\text{Spices Pack, Rubik's Cube}\} : C$

$\{\text{Classic Poster}\} : D$

$\{\text{Soap Bar}\} : E$

$\{\text{Language Textbook}\} : F$

$$\text{conf}(A \rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)} = \frac{2}{2} = 1$$

$\Rightarrow$ The laptop is always bought when Sci-Fi Book and Wireless mouse are bought.

$$\text{conf}(C \rightarrow D) = \frac{\text{support}(C \cup D)}{\text{support}(C)} = \frac{1}{2}$$

$\Rightarrow$ When a Spices Pack and Rubik's Cube are bought, there is a 50% chance that a Classic Poster is also bought.

$$\text{conf}(E \rightarrow F) = \frac{\text{support}(E \cup F)}{\text{support}(E)} = \frac{0}{2} = 0$$

⇒ The language Textbook is never bought when a Soap bar is bought.

(c) We assume the support threshold is $s=3$:

- **Step 1:** Let's denote the items we have:

"Laptop" = A, "Wireless Mouse" = B, "Sci-Fi Book" = C,

"Spices Pack" = D, "Classic Poster" = E, "Rubik's Cube" = F,

"Box of Chocolates" = G, "Horror Novel" = H, "Hammer" = I,

"Soap Bar" = J, "Anime Figurine" = K, "Replica Sword" = L,

"Comic Book" = M, "Language Textbook" = N.

- **Step 2:** Let's see how many times each one of these items appears in a transaction (count the support of every individual item):

$\text{supp}(A) = 5$, $\text{supp}(B) = 2$, $\text{supp}(C) = 3$, $\text{supp}(D) = 4$,

$\text{supp}(E) = 4$, $\text{supp}(F) = 2$, $\text{supp}(G) = 2$, $\text{supp}(H) = 2$,

$\text{supp}(I) = 2$, $\text{supp}(J) = 2$, $\text{supp}(K) = 3$, $\text{supp}(L) = 1$,

$\text{supp}(M) = 2$, $\text{supp}(N) = 2$

- **Step 3:** We eliminate items that has a support less than 3:

we will have only: A, C, D, E, K left.

- **Step 4:** Let's find all frequent pair (2-itemsets):

$\{A, C\}, \{A, D\}, \{A, E\}, \{A, K\}, \{C, D\}, \{C, E\}, \{C, K\},$
 $\{D, E\}, \{D, K\}, \{E, K\}$

let's count the support for these pairs:

$\text{supp } \{A, C\} = 3, \text{ supp } \{A, D\} = 0, \text{ supp } \{A, E\} = 0,$

$\text{supp } \{A, K\} = 2, \text{ supp } \{C, D\} = 0, \text{ supp } \{C, E\} = 0, \text{ supp } \{C, K\} = 0,$

$\text{supp } \{D, E\} = 2, \text{ supp } \{D, K\} = 1, \text{ supp } \{E, K\} = 0$

The only pair that hit the $s=3$ threshold is: $\{A, C\}$

→ Thus the frequent itemsets are: $\{A, C\}, \{A\}, \{C\}, \{D\}, \{E\},$
 $\{K\}$ ($\{\text{"Laptop"}, \text{"Sci-Fi Book"}\}, \{\text{"Laptop"}\}, \{\text{"Sci-Fi Book"}\},$
 $\{\text{"Spices Pack"}\}, \{\text{"Classic Poster"}\}, \{\text{"Anime Figurine"}\}$

(d) We assume we have the support threshold $s=4$: (iteration 1)

(We will use the same annotations and the support calculation we've already used in the previous task).

Candidate 1-itemsets: $C_1 = \{A, B, C, D, E, F, G, H, I, J, K, L, M, N\}$

- Now we eliminate the items that has a support less than 4.

We will have only: A, D, E ($\{\text{"Laptop"}, \text{"Spices Pack"}, \text{"Classic Poster"}\}$)

Frequent 1-itemsets: $L_1 = \{A, D, E\}$

Now in iteration we will find all frequent pair (2-itemsets).

Candidate 2-itemsets: $C_2 = \{A, D\}, \{A, E\}, \{D, E\}$

Let's count the support for these pairs.

$$\text{supp } \{A, D\} = 0, \text{ supp } \{A, E\} = 0, \text{ supp } \{D, E\} = 2$$

There is no pair that hit the $s=4$ threshold.

→ Thus the frequent itemsets are: $\{A\}, \{D\}, \{E\}$ ("Laptop", "Spices Pack", "Classic Poster")